open-cam: an open-source hyperspectral, ray-traced camera simulator built on pbrt-v4 for sensor, lens and CFA studies

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Abstract

We present open-cam, an open-source (MIT) camera simulator that couples the physically based renderer pbrt-v4 with a radiometrically explicit sensor model. Unlike most driving and robotics simulators, open-cam never forms an RGB image inside the renderer: pbrt-v4 writes per-wavelength radiance buckets, which are converted to sensor-plane irradiance, photon counts and photoelectrons with measured quantum efficiency and infrared-cut-filter curves, then passed through an EMVA1288-style noise, ADC and colour filter array model. Lenses can be traced through real prescriptions. We describe the architecture, compare it with open-source and commercial camera simulators, and report reproducible validation: an end-to-end ColorChecker render, a photon-transfer check that recovers the configured system gain to within 0.3%, a traced-lens slanted-edge MTF, and a study of non-Bayer filter arrays derived from a single spectral render. We are explicit about current limitations, notably that only Bayer mosaics are sampled and demosaiced end-to-end.

Introduction

Camera simulation is used to prototype sensors, to evaluate image signal processors (ISPs) and to generate training and test data for perception. Two families of tools exist. Image-systems simulators such as ISETCam [1, 2, 3] model the physics of light, optics and sensors in spectral units, while driving and robotics simulators (CARLA [4], AirSim [5], NVIDIA Isaac Sim [6] and commercial platforms) emphasise real-time scene dynamics and typically render RGB images to which camera effects are added afterwards.

Many questions that matter in automotive and mobile imaging cannot be answered from an RGB render: how a red–clear–clear–blue (RCCB) or red–yellow–yellow–cyan (RYYCy) mosaic changes low-light signal-to-noise ratio (SNR), how the infrared-cut filter (IRCF) interacts with an LED or tungsten illuminant, or how a lens with longitudinal chromatic aberration blurs each colour channel. All require the spectral radiance at the sensor. open-cam [7] is a Python toolkit built to answer such questions with free tools. Its contributions are: (i) a spectral pipeline from pbrt-v4 [8, 9] spectral film to photoelectrons, with every radiometric factor explicit; (ii) an EMVA1288-style [10] sensor model with analytic validators and real-renderer end-to-end tests; (iii) a library of camera recipes, measured spectral data and traced lens prescriptions; and (iv) an honest account of what is and is not yet implemented, in particular for non-Bayer colour filter arrays (CFAs).

Related work

Image-systems simulation. ISETCam (MATLAB, MIT licence) keeps scene spectral radiance, optical-image irradiance and sensor photoelectrons and voltages in physical units, and includes models of CFAs, optics and displays [3, 2]. ISET3D (MIT) adds 3D rendering with pbrt-v4 [11]; it has been validated against measured cameras [12, 13] and extended with ray-transfer functions for lenses whose design is hidden [14]. ISETAuto (Apache-2.0) applies the stack to driving scenes and network evaluation [15, 16, 17]. open-cam is closest to this family; it differs in being pure Python, in its configuration-driven catalogue of 73 camera recipes, and in its EMVA1288-oriented validators and continuous-integration tests that run the real renderer. It does not yet offer the breadth of ISETCam's optics, display and human-vision models.

Renderers. pbrt-v4 (Apache-2.0) renders with spectral sampling and provides a `spectral` film that stores radiance in wavelength buckets, as well as a `realistic` camera that traces a lens prescription [8, 18, 19]. Mitsuba 3 (BSD-style licence) offers spectral and polarised variants [20, 21]. Both are renderers, not camera models: they stop at radiance or at a colour image.

Driving and robotics simulators. CARLA (MIT) renders RGB cameras with Unreal Engine post-processing; its documented camera attributes include auto exposure, bloom, grain jitter and lens-distortion parameters [22]. AirSim (MIT) exposes noise and distortion settings for its RGB cameras [23]; Gazebo models camera noise as independent Gaussian noise per pixel [24]. NVIDIA Isaac Sim's RTX camera documents OpenCV pinhole and fisheye distortion models, additive noise, encoding of the RGB image into a single-channel Bayer mosaic and an on-chip ISP stage [6]; NVIDIA's DRIVE simulation stack advertises high-fidelity sensor simulation [25]. None of these documentations describe spectral rendering into a QE-weighted sensor.

Commercial sensor and optics tools. Ansys Speos performs spectral, wavelength-dependent ray tracing with camera and detector response simulation [26, 27]; Ansys AVxcelerate Sensors provides a real-time, physics-based camera model parameterised from a camera datasheet and a multispectral material database [28]. Lens-design packages (Ansys Zemax OpticStudio [29], Synopsys LightTools [30]) provide sequential and non-sequential ray tracing and stray-light analysis. Scenario-level platforms, including Siemens Simcenter Prescan [31], IPG CarMaker [32], Hexagon VTD [33], rFpro [34], Cognata [35] and dSPACE AURELION, describe "physics-based" or ray-traced sensor simulation; rFpro, for example, documents a multi-exposure HDR camera API with motion blur [34]. Foretellix [36] targets scenario generation and coverage rather than sensor physics. For these commercial tools the spectral resolution, noise model and CFA options are not specified in the public material we could access, so Table 1 marks them as not publicly documented rather than guessing. Pricing is quote-based throughout.

Measurement. EMVA1288 [10, 37], photon transfer [38] and ISO 12233 slanted-edge analysis [39, 40] define how real cameras are characterised; the EMVA1288 reference implementation (LGPL-3.0) [41] and commercial suites such as Imatest [42] implement them for measured data. open-cam uses the same definitions so that simulated and measured cameras can be compared on equal terms.

Camera-simulation capabilities as stated in each tool's public documentation or papers (accessed October 2026). "n.d." = not documented in the public sources we reviewed; it does not mean the capability is absent.

Tool	Spectral rendering	Ray-traced lens	Sensor noise / EMVA1288	CFA flexibility	Licence / cost
open-cam	Yes: pbrt-v4 spectral film, user-set buckets	Yes: pbrt-v4 realistic camera, 4 prescriptions	Shot, read, PRNU, DSNU, dark current, ADC, FPN, defects; EMVA validators	Arbitrary QE curves; Bayer (4 phases) end-to-end only	MIT; free
ISETCam / ISET3D / ISETAuto	Yes (spectral radiance)	Yes (pbrt-v4; ray-transfer functions)	Physical photoelectron/voltage model	CFA models included [3]	MIT / Apache-2.0; MATLAB required
pbrt-v4, Mitsuba 3	Yes	pbrt-v4: realistic camera	None (renderer)	None	Apache-2.0 / BSD; free
CARLA, AirSim, Gazebo	No (RGB)	No; distortion parameters	Grain / Gaussian / noise-line effects	None (RGB output)	MIT / Apache-2.0; free
Isaac Sim (RTX)	n.d.	No; OpenCV distortion models	Additive noise	Bayer encoding + ISP	NVIDIA licence
Ansys Speos	Yes	Yes	Detector response, raw signal	n.d.	Commercial
Ansys AVxcelerate	Multispectral materials	n.d.	Datasheet-parameterised	n.d.	Commercial
Zemax OpticStudio, LightTools	Wavelength-dependent tracing	Yes (lens design)	n.d.	n.d.	Commercial
Prescan, CarMaker, VTD, rFpro, Cognata, AURELION, DRIVE Sim	n.d.	n.d. (rFpro: ray tracing)	n.d.	n.d.	Commercial
Imatest, emva1288	– (measurement)	–	EMVA1288 analysis of measured data	–	Commercial / LGPL-3.0

Architecture

open-cam is a set of command-line tools orchestrated by `tools/run_pipeline.py`. A single YAML file (`config/pipeline.yaml`) selects a camera recipe and drives four stages: scene generation, pbrt-v4 rendering, spectral-to-electron conversion and the sensor/noise/CFA stage, followed by validators. A recipe in `config/camera_recipes/` (73 are provided: phone, compact, interchangeable-lens and research cameras) references a sensor model (QE curves, pixel pitch, fill factor, exposure, EMVA parameters, CFA) and a lens model. Sensor models are written as deltas from `config/sensor_models/default.yaml`, so a new sensor needs only its overrides.

Scenes. `build_colorchecker_scene.py` builds a 24-patch ColorChecker from measured X-Rite reflectances lit by a tabulated illuminant (D50–D75, A, C, fluorescent and LED spectra are included). `build_image_quality_targets.py` generates slanted-edge, Siemens-star and ISO-style noise targets, and `build_munsell_scenes.py` builds charts from 1269 measured Munsell chips (380–800 nm at 1 nm).

Spectral rendering. Scenes are rendered with pbrt-v4's `spectral` film, which writes an OpenEXR file with one radiance channel per wavelength bucket (`S0.`$\langle\lambda\rangle$`nm`, in $\mathrm{W\,m^{-2}\,sr^{-1}\,nm^{-1}}$) [18]. The default pipeline uses 64 buckets between 360 and 830 nm. Cameras can be pinhole, thin-lens or pbrt-v4's `realistic` camera, which traces rays through a lens prescription [19]. The repository ships four prescriptions (a 22 mm wide-angle, a 50 mm double-Gauss, a 250 mm telephoto and a 10 mm fisheye) and 14 lens models that bind them, or thin-lens parameters, to camera classes.

Radiance to electrons. Radiance is converted to sensor-plane irradiance by `pbrt_spectral_exr_to_electrons.py` using the camera equation

$$E_\lambda(x,y) = L_\lambda(x,y)\,\frac{\pi\,\tau(\lambda)}{4N^2(1+m)^2}\,c_{\mathrm{cal}}, \tag{1}$$

where N is the f-number (taken from the traced prescription when the realistic camera is used), m the magnification, τ the optics transmittance and c_{cal} an optional photometric calibration to a target scene illuminance. The mean electrons in channel c are then

$$\mu_{e,c} = \sum_\lambda E_\lambda \frac{\lambda}{hc}\,\eta_c(\lambda)\,T_{\mathrm{IR}}(\lambda)\,\Delta\lambda\;A_{\mathrm{px}}\,t\,F\,V_c(x,y), \tag{2}$$

with QE η_c, IRCF transmittance T_{IR}, bucket width $\Delta\lambda$, pixel area A_{px}, exposure time t, fill factor F and an optional vignetting map V_c. The photon conversion, QE, bin widths and geometry are folded into a $K \times C$ weight matrix (`sensor_radiometry.spectral_electron_weights`), so integration is a single matrix product per pixel. The same helper is shared with `spectral_sensor_forward.py`, an analytic forward model used for fast tests. QE curves for many cameras are taken from a measured spectral-sensitivity database; generic red, green, blue, cyan, yellow, magenta and monochrome curves and an IRCF are also provided.

Sensor model. `apply_emva_noise.py` implements the EMVA1288 linear camera model [10]: Poisson shot noise on signal plus dark current (with a temperature model), PRNU as a multiplicative gain map, DSNU as an additive offset map, Gaussian read noise, optional kTC noise, row/column fixed-pattern and flicker noise, hot/stuck pixels, blooming, inter-pixel crosstalk, full-well clipping and an ADC with gain K ($\mathrm{e^-}$/DN), black level, bit depth and optional

INL/DNL. Spatial maps use a fixed seed (one "camera unit") while temporal noise uses a per-frame seed. A Bayer mosaic is sampled with any of the four RGGB/BGGR/GRBG/GBRG phases and demosaiced bilinearly or with the Malvar–He–Cutler algorithm [43]. `apply_spectral_psf.py` adds post-render optics where a traced lens is not used: Gaussian, wavelength-dependent Gaussian and Airy-disk PSFs per spectral bucket, lateral chromatic aberration and stray light.

Validation and tests. `validate_emva_model.py` compares the analytic EMVA mean and variance with Monte Carlo at photon-transfer levels and checks the recipe against its datasheet block; `batch_validate_emva.py` runs this across all recipes. `validate_demosaic_linear.py` checks demosaic fidelity in linear DN. Twenty test modules include two that call the real pbrt-v4 binary: `test_pbrt_e2e.py` and `test_pbrt_full_pipeline.py`. The latter renders a 96×64, 16-bucket ColorChecker and, for every pixel, forms the residual $z = (\mathrm{DN} - (b + \mu_e/K))/\sigma_{\mathrm{pred}}$ against the analytic EMVA prediction, requiring zero mean and unit variance; it also checks ADC limits, seed determinism and demosaic error.

Teaching. A Dear PyGui [44] desktop application (`gui/`) exposes the same engines through interactive scenarios on optics, PSF and aberrations, MTF, sensor noise, exposure and defects, colour/ISP and image generation, with accompanying tutorials.

Results

All results below were produced on a CPU build of pbrt-v4 with the scripts in `paper/ei2027/scripts/`; the exact commands are listed in the accompanying README. Renders are 960×640.

End-to-end ColorChecker. We rendered the ColorChecker with the geometry and lens of `config/pipeline.yaml` (realistic 22 mm wide-angle prescription, 8.756 mm aperture, focus 4 m, 1024 samples per pixel, 64 buckets over 360–830 nm) under CIE D65 and CIE A, and ran it through the `iphone_8` recipe (1.4 µm pixels, GBRG, Malvar demosaic, $K = 9.906\,\mathrm{e^-/DN}$, $9500\,\mathrm{e^-}$ full well, 10-bit) at 500 lux and 0.12 s, the pipeline defaults. With the realistic camera the effective f-number from the prescription is f/2.51. The white patch yields 364/821/439 $\mathrm{e^-}$ (R/G/B) under D65. Fig. 1 shows the demosaiced output with grey-world preview white balance; the gains change from 2.19/1.00/1.83 (D65) to 1.27/1.00/3.21 (A), a direct consequence of integrating each illuminant's spectrum against the camera QE rather than of an RGB tint. Against a noise-free reference, `validate_demosaic_linear.py` reports a Malvar demosaic error of 0.114 DN MAE and 0.389 DN RMSE.

Photon transfer. Using the iphone_8 EMVA parameters ($\sigma_d = 1.6\,\mathrm{e^-}$, PRNU 1.8%, DSNU $0.16\,\mathrm{e^-}$), we simulated 256×256 uniform frame pairs at 29 exposure levels from dark to beyond saturation with `emva_theory.simulate_uniform_stack`, and applied the EMVA1288 photon-transfer method (temporal variance from frame-pair differences; zero-intercept fit of $\sigma_y^2 - \sigma_{y,\mathrm{dark}}^2$ against $\mu_y - \mu_{y,\mathrm{dark}}$ up to 70% of full well). The fit recovers $K = 9.883\,\mathrm{e^-/DN}$ against the configured 9.906 (0.23% error) and $\sigma_d = 1.588\,\mathrm{e^-}$ (configured 1.6). The EMVA PRNU estimate from 16-frame dark and 50%-saturation stacks is 1.79% (configured 1.8%). Across the linear range the simulated variance deviates from the analytic model by at most 1.3% (median 0.46%), consistent with the sampling error of a 65,536-pixel estimate (Fig. 2). `validate_emva_model.py` passes all photon-transfer, dark-frame and datasheet checks for this recipe with 20,000 Monte Carlo trials.

Traced-lens MTF. We rendered a 5° slanted edge at 3.2 m through the 50 mm double-Gauss prescription (32 buckets, 400–700 nm, 4096 samples per pixel) at f/2 in focus, at f/2 focused at 2.6 m, and at f/5.6 focused at 2.6 m, and analysed it with `sfr_analysis.slanted_edge_sfr`. With the generic green QE the MTF50 is 0.257, 0.328 and 0.340 cy/px respectively, and the in-focus MTF at Nyquist is 0.267 (Fig. 3). Two observations are worth reporting as measured rather than explained away. First, the nominal `focusdistance` of 3.2 m does not maximise MTF50: focusing at 2.6 m gives a higher MTF50, but the two f/2 curves cross near 0.4 cy/px and the nominal focus keeps more contrast towards Nyquist (0.267 against 0.191). This is consistent with residual spherical aberration at f/2, where moving focus trades a sharp core for a smaller halo, though we have not verified that cause. Best focus therefore depends on the frequency of interest and should be found with a through-focus sweep, not the nominal value. Second, stopping down to f/5.6 raises MTF50 further, as expected from the smaller blur circle. Because the film is spectral, the same render also yields an MTF per wavelength; Fig. 3 (right) shows MTF50 in bands of four buckets (37.5 nm). open-cam traces the glass dispersion of the prescription (the patent's ν_d values, with the $n(\lambda)$ shape of the nearest SCHOTT glass), and the bands show the resulting axial colour. At f/2 in focus, MTF50 is highest near 490–530 nm and falls towards both ends of the band (0.18–0.27 cy/px); at f/5.6 focused at 2.6 m the curve is U-shaped (0.33–0.46 cy/px), consistent with blue and red foci lying closer to the edge distance than the green focus. With dispersion switched off (`-no-lens-dispersion`) the bands are flat to within render noise, so the variation comes from the glass model, not the renderer. An earlier version of this analysis showed a periodic ripple across buckets; it was a pbrt-v4 wavelength-sampling artefact (one wavelength offset shared by the four hero wavelengths, and a seed override), which open-cam now removes by jittering each wavelength stratum independently. Edge positions were located with a 50%-crossing fit after per-row flat-fielding, because the derivative-centroid estimator in `sfr_analysis.py` was biased by render noise on this edge (it reported a −2.3° edge for a measured −5.0°); the ESF binning, LSF and MTF use the repository functions. These values are in film pixels and include the pixel aperture: each pixel integrates radiance over its own area (a box reconstruction filter one pixel wide), whose MTF $|\mathrm{sinc}\, f|$ is 0.64 at Nyquist. Dividing it out gives a lens-only in-focus MTF at Nyquist of 0.42.

CFA support: what exists. We inspected the CFA path to state its scope precisely. `apply_emva_noise.py` samples and demosaics only 2×2 Bayer phases and rejects other pattern names. The recipes named `default_rccb`, `default_ryycy`, `default_rccg`, `default_rggcy` and `default_cmy` are *RGB proxies*: the three QE slots of the Bayer path are filled with other curves (for RCCB, the "clear" channel is the generic cyan curve in the green slot; for RYYCy, yellow and cyan replace green and blue), and the result is still sampled on a Bayer lattice. RGBW, true clear pixels, quad-Bayer [45] and arbitrary $n \times n$ tiles are therefore not yet supported end-to-end, and there is no non-Bayer demosaicing.

CFA support: what the spectral design enables. Because the renderer output is spectral, any channel set can nonetheless be computed from the same EXR with the pipeline's own integration weights, and sampled on any tile. `fig_cfa.py` does this outside the pipeline (no pipeline code is modified): it integrates the D65 and A ColorChecker renders against six QE×IRCF curves (Fig. 4), with "W" the unfiltered monochrome silicon QE, and anchors the absolute scale to the pipeline's green-channel electrons for the default recipe

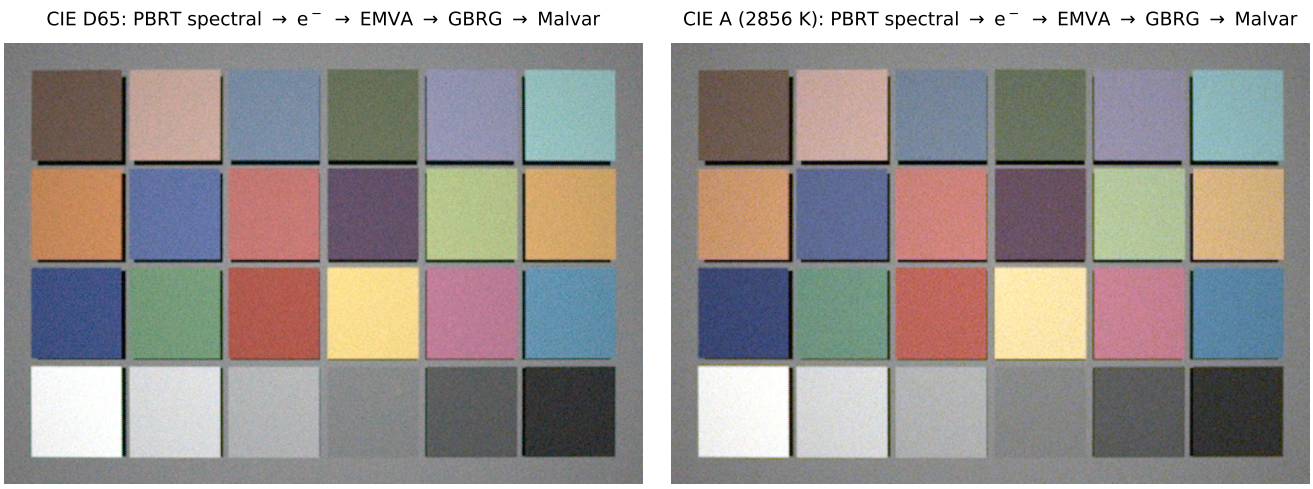


Figure 1. *ColorChecker rendered spectrally by pbrt-v4 (64 buckets) and processed by open-cam with the iphone_8 recipe: electrons → EMVA noise → GBRG mosaic → Malvar demosaic, grey-world preview white balance, no colour correction. Left: D65; right: CIE A.*

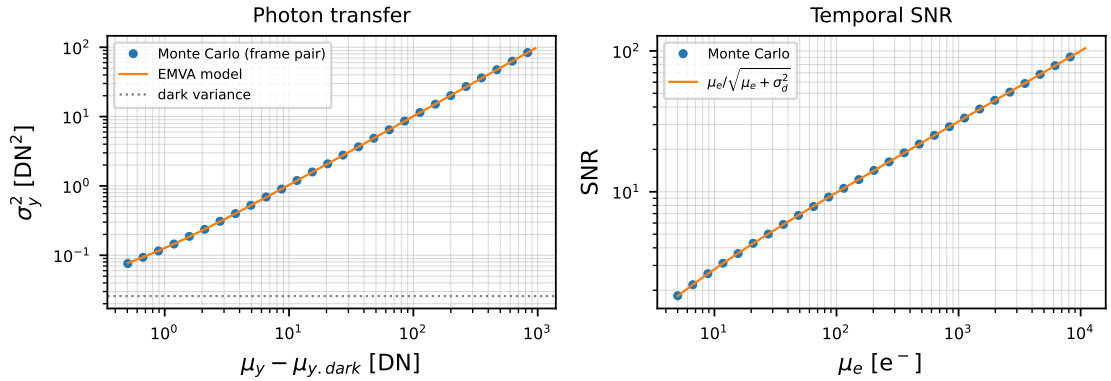


Figure 2. *EMVA1288 photon-transfer validation for the iphone_8 recipe. Left: temporal variance versus dark-subtracted mean signal for Monte Carlo frame pairs (markers) and the analytic model (line); the gain fit uses levels up to 70% of full well. Right: temporal SNR versus mean electrons.*

(the ratio is constant across all 24 patches to a coefficient of variation of 4×10^{-8}, confirming that the script and the pipeline implement the same radiometry). Fig. 5 shows raw mosaics for six layouts. Table 2 reports, for the neutral 5 patch (about 19% reflectance), the luminance-channel electrons at 500 lux and the analytic EMVA temporal SNR, $\mu_e/\sqrt{\mu_e + \sigma_d^2}$ with $\sigma_d = 2.6\,\mathrm{e}^-$, scaled to 10 lux. A clear pixel collects 2.62× the electrons of the green pixel and gains 6.0 dB of SNR in this low-light case; yellow and the cyan proxy gain 2.8 and 3.2 dB. These ratios depend on the illuminant: under CIE A the yellow/green ratio rises from 1.55 to 2.12 and the cyan/green ratio falls from 1.65 to 1.11. The spectral render also quantifies the error of the current RGB proxy: normalised on the white patch, the cyan-proxy/clear ratio ranges from 0.43 (patch 15, red) to 1.34 (patch 13, blue), so a cyan curve is not a scalar stand-in for a clear pixel, and the error depends on scene colour. Quad-Bayer has identical per-pixel statistics to Bayer here because binning and remosaicing are not modelled. Per-pixel analyses of this kind are impossible from an RGB render, which is why we regard the spectral representation, rather than the current CFA code, as open-cam's main asset for CFA research [46].

Luminance-channel signal on the neutral 5 patch from one D65 render (default recipe, 0.12 s) and analytic temporal SNR at 10 lux.

CFA (channel)	μ_e @500 lx	SNR @10 lx	Δ	ch/G (A)
Bayer / quad (G)	282	4.1 dB	–	1.00
RGBW, RCCB (W)	741	10.1 dB	+6.0 dB	2.67
RCCB proxy (Cy)	465	7.3 dB	+3.2 dB	1.11
RYYCy (Ye)	438	6.9 dB	+2.8 dB	2.12

Discussion and limitations

The results support the central design choice: keeping light spectral until the photon-counting step makes illuminant, IRCF and filter questions first-class, and the EMVA validators make the noise model auditable against datasheets. The limitations are equally concrete. Non-Bayer CFAs are not sampled or demosaiced in the pipeline; the natural extension is to generalise the Bayer lookup tables in `apply_emva_noise.py` to an $n \times n$ tile of named QE curves with

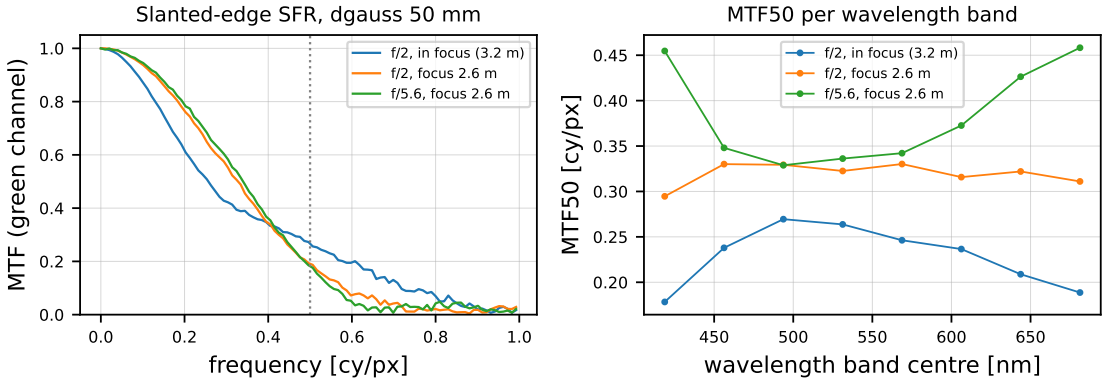


Figure 3. *Slanted-edge MTF through the traced 50 mm double-Gauss lens. Left: green-channel MTF for three aperture/focus settings (dotted line: Nyquist). Right: MTF50 of the same renders in 37.5 nm wavelength bands; the variation is the axial colour of the traced glasses (see text).*

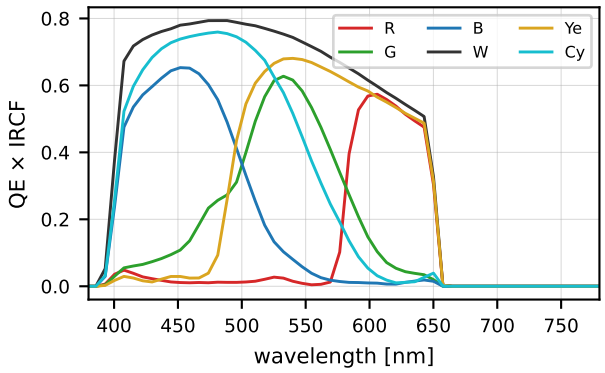


Figure 4. *Generic QE×IRCF curves shipped in* `spectra/QE/interpolated/` *and used for the CFA study (W: unfiltered monochrome QE).*

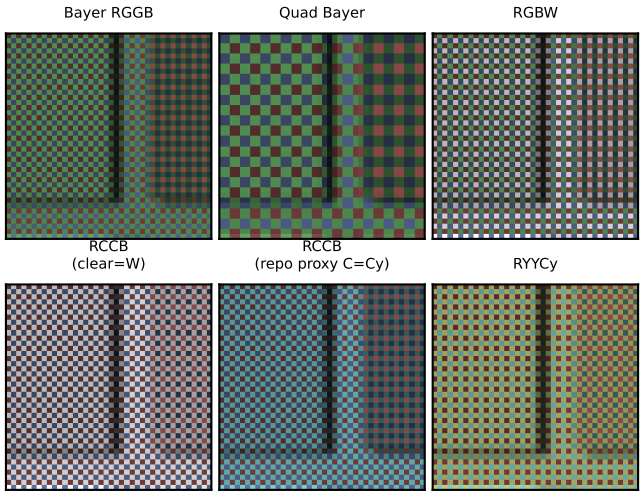


Figure 5. *Raw mosaics computed from one spectral render (D65) at the corner of patches 14, 15, 20 and 21, each pixel tinted by its filter. Only Bayer is sampled by the pipeline itself; the other layouts are produced by* `fig_cfa.py` *from the same spectral buckets.*

per-tile demosaicing, which the spectral integration already supports. Many recipes carry EMVA parameters inferred from public tables (each recipe records a calibration tier) rather than measured reports. Rendering is offline: the 1024-sample, 64-bucket ColorChecker took roughly 4.5 min on an 8-core CPU, so open-cam complements rather than replaces real-time simulators. Scenes are currently charts and targets rather than driving environments, and there is no motion blur, rolling shutter or HDR multi-exposure model yet.

Conclusion

open-cam provides a free, scriptable path from pbrt-v4 spectral radiance to EMVA-consistent raw data through traced lenses and measured QE curves. It recovers configured sensor parameters through the EMVA1288 photon-transfer method, reproduces illuminant-dependent white balance from first principles, and, while its pipeline CFA support is currently Bayer-only, its spectral representation already allows quantitative studies of RGBW, RCCB and RYYCy designs. Code, configurations and all scripts used for this paper are available at `https://github.com/briandeegan82/open-cam`.

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